COMPUTER VISION PROJECT

Smart Attendance System

— Face Recognition Technology —

2026

01 Project Overview

Project Title	Smart Attendance System using Face Recognition
Objective	Automatically detect and recognize human faces to record attendance
Input	Uploaded images — reference photos + test image
CV Technique	Face Detection + Recognition (128-dim dlib encoding)
Output	Annotated image + CSV file + HTML Report
Libraries	OpenCV, face_recognition, NumPy, Matplotlib, CSV, datetime
Platform	Google Colab — Python 3

02 Problem Statement

Traditional attendance systems rely on manual roll calls or ID cards — time-consuming and prone to human error. This project proposes an automated Computer Vision solution that identifies individuals from images and logs attendance instantly.

Key challenges addressed:

- Eliminating manual attendance marking entirely.
- Accurately identifying faces under varying lighting conditions.
- Generating clean, structured attendance records automatically.

03 Preprocessing Steps

Step	Method	Purpose
1	Resize to 50%	Speeds up face detection without quality loss
2	BGR → RGB	Required color space for face_recognition library
3	Normalize 600×600	Consistent input size for encoding reference images

04 Feature Extraction & Detection

The system uses face_recognition (built on dlib) which applies a deep neural network model to extract and compare facial features:

- Detect face locations using the HOG (Histogram of Oriented Gradients) model.
- Extract a 128-dimension feature vector that uniquely represents each face.
- Compare detected encodings against all known encodings using Euclidean distance.
- Threshold of 0.5 — distances below this are considered a confirmed match.

05 Output & Results

Output File	Description
Annotated Image	Bounding boxes with name and confidence % drawn on each detected face
CSV File	Attendance log: Name, Date, Time, Confidence, Status
HTML Report	Professional dashboard with stats cards and embedded result image

06 User Interaction

- File upload widgets to provide reference photos and test images via Colab.
- Step-by-step notebook cells guide the user through each stage clearly.
- Real-time printed feedback showing detected names and confidence scores.
- Automatic download at the end: CSV file, annotated image, and HTML report.